IST 691: Research Methodology and Scientific Writing Skills)

The Experimental Research Approach

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Outline

- Causation
- What is an Experiment
- Advantages and Disadvantages of the Experimental Approach
- Experimental research settings
  - Field experimentation
  - Laboratory experimentation

Causation

- Causation (meaning)
  - Manipulation of one event causes another event
- Experimental research approach
  - Scientific approach to research
    - One or more independent variables are manipulated and their impact on dependent variables observed to measure their effect of the former on the latter
    - i.e., to identify cause-and-effect relationships among dependent and independent variables

Causation

- Dealing with causation can be tricky
  - Assume we beat a child for colouring the wall and he stops colouring the wall
  - Why did the child stop colouring the wall?
    - The beating?
    - Because he got tired of colouring the wall?
    - Because his attention was drawn to other interests?

Example: History of Attempts to Find the Cause of Malaria

- Mediaval times
  - The disease was common in lowlands and was thought to be as a result of the bad air at the lowlands
  - As a result, towns were built at hill tops
  - But the disease did not disappear, but its incidence and severity were reduced to a level consistent with productive community life

Example: History of Attempts to Find the Cause of Malaria

- Introduction of quinine in Europe
  - Apparently, quinine acted on some situation within the patient to relieve and often cure him completely
  - Also, the malaria parasite was discovered in the blood of patient suffering from the disease
  - The effectiveness of quinine was explained by its ability to eliminate this parasite from the blood
  - The malaria parasite was then deemed to be the cause of malaria
  - But malaria did not disappear as a public health problem

Example: History of Attempts to Find the Cause of Malaria

- Discovery that the malaria parasite was transmitted by certain species of mosquitoes
  - The cause of the disease became the mosquito, and attention was directed to control its activities
  - Entertainingly, malaria has now disappeared from large parts of the world without anyone doing much about it
  - Many areas in the rich North still produce mosquitoes capable of transmitting the parasite, and people carrying the organism still come to these areas from time to time, but the disease is no longer in these areas
    - Perhaps the epidemic of malaria is the result of a nicely balanced set of social , economic, and biological factors, each at an appropriate level

Necessary and Sufficient Condition

- As the malaria story illustrates, it is hard to identify a single factor that causes an effect
- Identification of causation can only occur when no alternative interpretations of an effect exist other than the one specified
  - At this stage, we have identified both the necessary and sufficient conditions for the occurrence of an event

Necessary and Sufficient Condition

- Necessary condition
  - A condition that must be present in order for an effect to occur, but it isn't enough on its own
    - To become an alcoholic, you must consume alcohol
    - Oxygen is necessary for fire
- Sufficient condition
  - A condition that will always produce an effect
  - Guarantees the event will happen if it is present but it might not be the only way
    - Destroying the auditory nerve always results in a loss of hearing, but is not the only way
    - Breaking a leg is sufficient for pain, but is not the only way for pain to be felt

Necessary and Sufficient Condition

- Key difference
  - Necessary: Required, but not enough.
  - Sufficient: Enough, but not necessarily required (other ways might exist).
  - Necessary & Sufficient: The only way and guaranteed way
- A condition must be both necessary and sufficient to qualify as a cause
  - The effect would never occur unless the condition were present
  - Whenever the condition was present, the effect would occur

Necessary and Sufficient Condition

- If a condition were only sufficient, the effect could occur in other ways
  - There are several ways one could lose hearing other than destruction of the auditory nerve
- If a condition were only necessary, then the effect may not occur
  - All people who consume alcohol do not become alcoholics, but one must consume alcohol in order to become an alcoholic
- Once the cause of an event is found (necessary and sufficient condition), a complete explanation for the occurrence of the event has been isolated
  - Unlike the malaria example, the explanation *will never change*

Necessary and Sufficient Condition

- It may be hard to find necessary and sufficient conditions, even if several studies suggest them
- One point of view holds that
  - Confirmation of an experiment only states that the hypothesis we were testing has survived the test
  - On the other hand, if the experiment fails to confirm a prediction or theory, the prediction or theory being tested is falsified (position of falsification)
    - Position of falsification
      - The belief that the best that can be said about a theory or prediction is that it is *not yet falsified*

Necessary and Sufficient Condition

- Another point of view
  - Sees causation as a large network of cause-and-effect relations
  - Hence in the malaria example, covariation was found between a number of events and malaria, each of which was once considered a specific cause-and-effect
    - Bad air in lowlands was found to covary with incidence of malaria
    - Presence of a parasite was found to covary with incidence of malaria
    - A certain species of malaria parasite-carrying mosquito was found to covary with incidence of malaria
  - This school of thought holds that for an effect to occur, a nicely balanced system of interrelated conditions must exist
    - For malaria to occur, the mosquitoes and parasites must exist in a system of other specific social and economic conditions

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What is an Experiment

- An objective observation of phenomena which are made to occur in a strictly controlled situation in which one or more factors are varied and the others are kept constant
- Designed specifically to answer the question of whether there is a causal relationship between two variables
  - Whether changes in an independent variable cause a change in a dependent variable

What is an Experiment

- Two fundamental features in an experiment
  - Researcher manipulates, or systematically varies, the level of the independent variable
    - The different levels of the independent variable are called conditions
  - Researcher controls, i.e., minimizes the variability in, variables other than the independent and dependent variable
    - These other variables are called extraneous variables

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Advantages of the Experimental Approach

- Inferring causal relationship
- Ability to manipulate precisely, one or more variables of the experimenter's choosing
- Usefulness – has produced results that have
  - Lasted over time
  - Suggested new studies
  - Suggested solutions to practical problems

Disadvantages (Criticisms) of the Experimental Approach

- (Laboratory) findings are obtained in laboratory conditions (artificial and sterile atmosphere) that are very different from real-life situations
- Difficulties in designing the experiment and the long time it may take

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Experimental Research Settings

- Field Experimentation
  - Experiments carried out in a natural setting (i.e., outside of laboratory settings)
  - Not subject to the artificiality problem
  - However, control of extraneous variables cannot be as good as in laboratory experiments
- Laboratory Experimentation
  - Experiment carried out in a laboratory setting
  - Experimenter can control and eliminate influence of extraneous variable
  - Price of this level of control however, is the artificiality of the situation created